

The Mathematics of a Trading System

Expectancy, variance, the risk/reward tradeoff and the risk of ruin — measured, not assumed, on real OHLCV.

A walk through what a rules-based backtester actually proves once costs, position sizing and luck are taken seriously. Strategy: **ma_crossover**. Instruments: **SPY, QQQ, PETR4.SA, VALE3.SA, ITUB4.SA**.

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1. Method — where the numbers come from

The win rate, expectancy and risk figures in this report are **not downloaded**. Yahoo Finance supplies only OHLCV; everything else is a *result* of simulating the strategy trade by trade. The engine is built to avoid the errors that flatter a backtest:

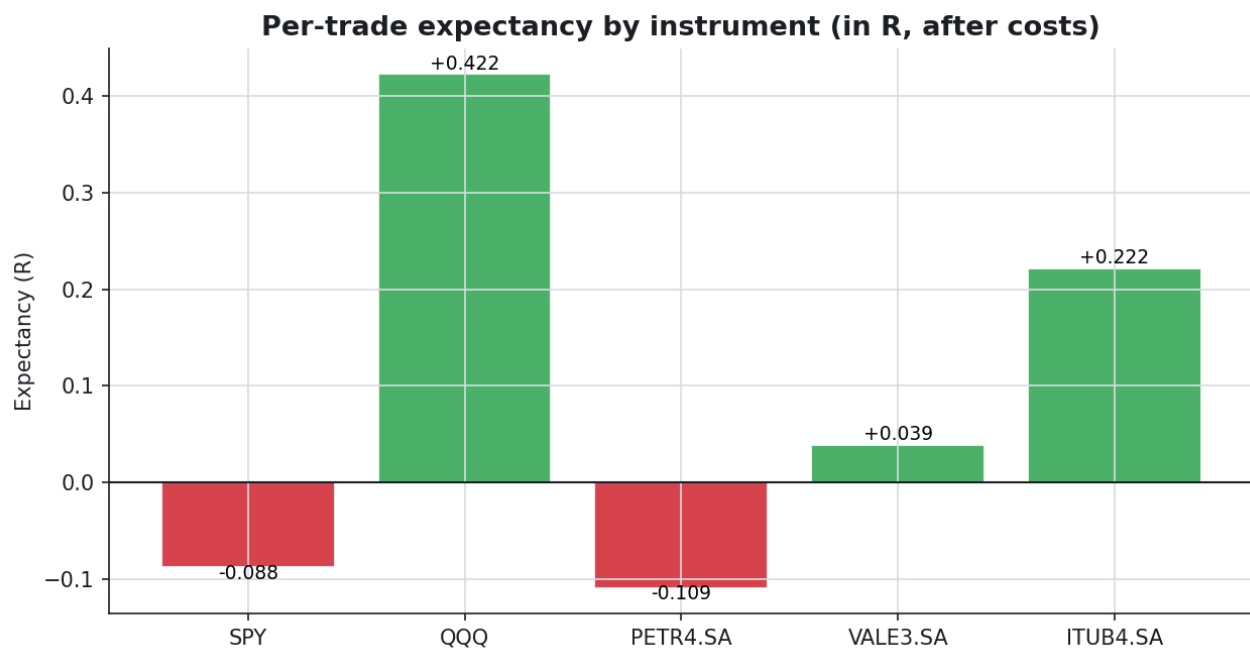
- **No lookahead.** A signal at the close of bar t is executed at the open of bar $t+1$ — never at the price that produced it.
- **Costs on every trade.** Spread, commission and slippage are charged round-trip; without them, paper expectancy lies.
- **Fixed-fractional risk.** Each trade risks a constant percentage of equity; a wider stop means a smaller position.
- **Conservative fills.** If a bar touches both stop and target, the stop is assumed to fill first.
- **Reproducible variance.** The Monte-Carlo bootstrap uses a fixed seed, so every run is identical.

Result in R (the R -multiple) is the unit that connects everything: a system is described by its win rate and its average R . A loss at the stop is $-1R$; the breakeven win rate, $1/(R+1)$, is the bar the real win rate must clear.

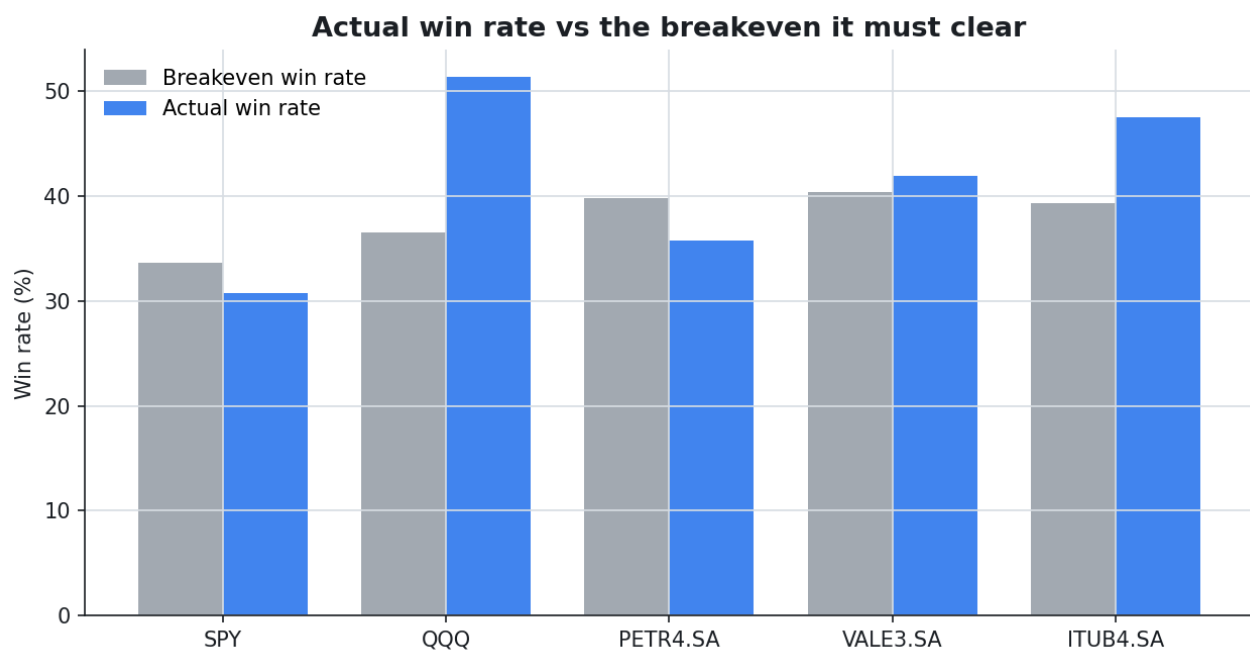
2. Cross-instrument scorecard

The same strategy, same parameters, run on every instrument. Where the expectancy is positive (after costs) the edge survives; where it is negative the geometry is just folklore on that market.

Instrument	Trades	Win%	Exp. R	Exp. \$	Profit factor	Max DD%	Edge?
SPY	39	30.8	-0.088	-4.61	0.87	5.6	no
QQQ	37	51.4	+0.422	+21.63	1.83	1.9	YES
PETR4.SA	42	35.7	-0.109	-5.63	0.84	3.7	no
VALE3.SA	43	41.9	+0.039	+1.70	1.06	3.6	YES
ITUB4.SA	40	47.5	+0.222	+11.09	1.40	4.7	YES



Per-trade expectancy in R, after costs. Green bars have a positive edge.



Actual win rate against the breakeven it must beat. A bar to the right of its grey twin is profitable.

3. SPY — the full fingerprint

Metric	Value	Metric	Value
Trades	39	Profit factor	0.87
Win rate	30.77%	Payoff (R)	1.98
Avg win	101.45 (+2.06R)	Avg loss	-51.76 (-1.04R)
Expectancy \$	-4.61	Expectancy R	-0.088
Breakeven WR	33.59%	Actual WR	30.77%
Total return	-1.80%	CAGR	-0.12%
Max drawdown	5.58%	Sharpe / Sortino	-0.10 / -6.90

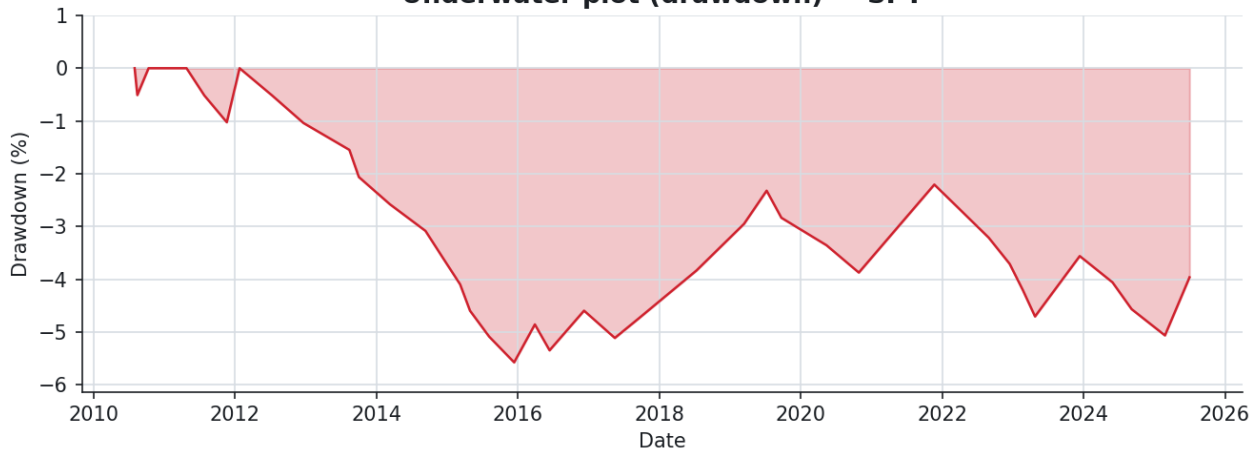
■ Small sample (39 trades < 100): expectancy is still noise-dominated. Treat with skepticism.

Equity curve — SPY



Realized equity curve, costs included.

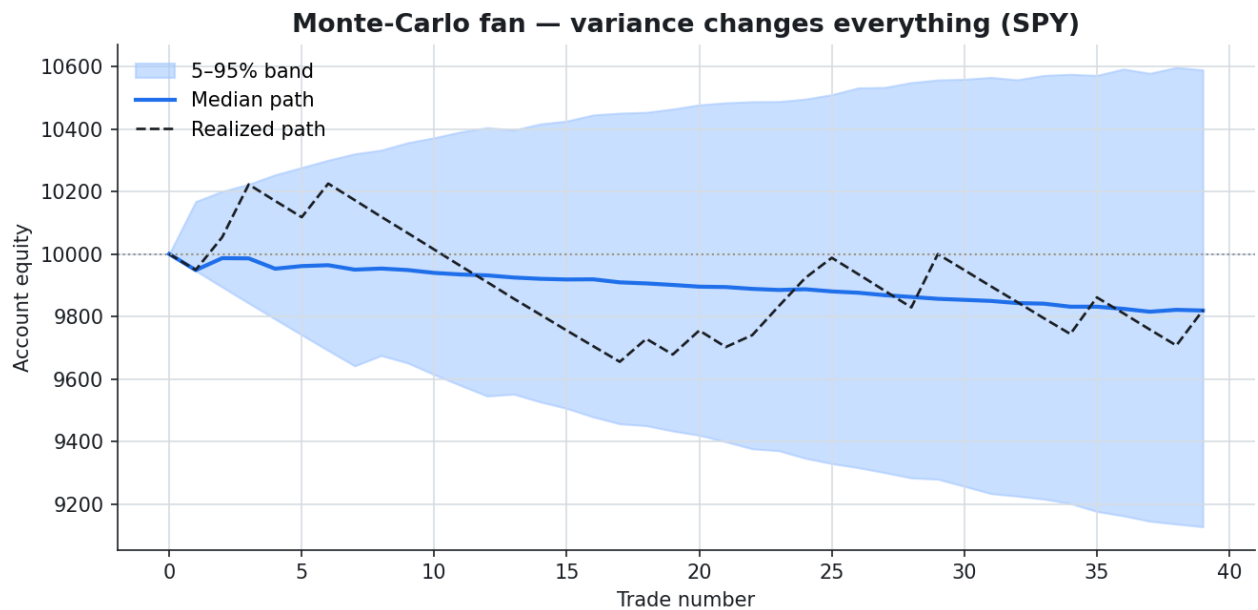
Underwater plot (drawdown) — SPY



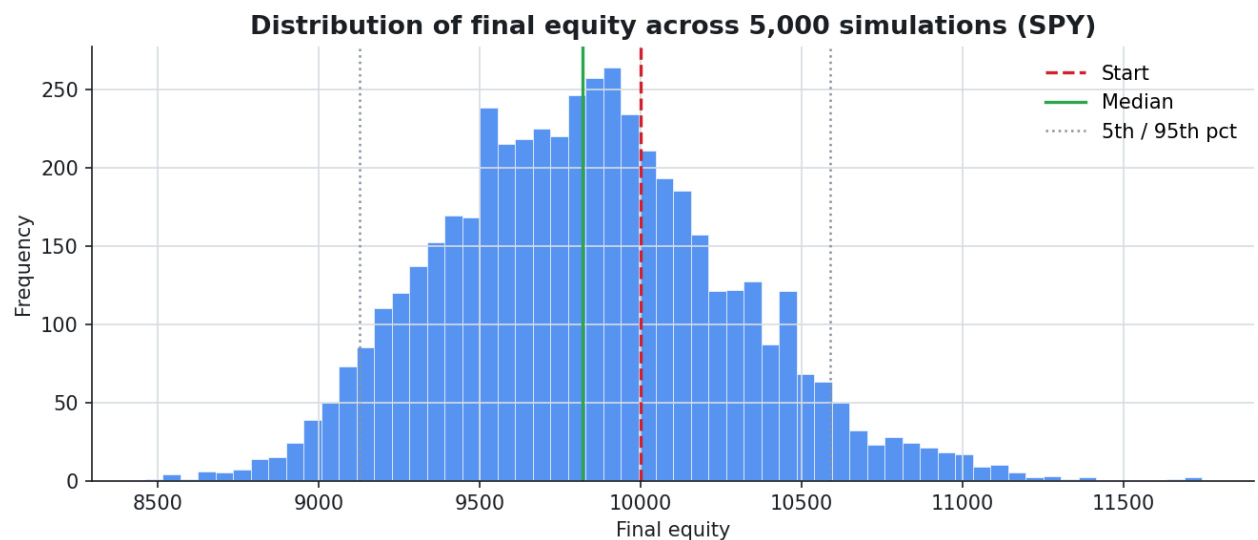
Underwater plot: time spent below the previous equity peak.

3.1 Variance — same edge, different luck

Reshuffling the realized trades 5,000 times shows the range of outcomes the same system could have produced. The median is the honest expectation; the 5–95% band is the experience you must be able to sit through.

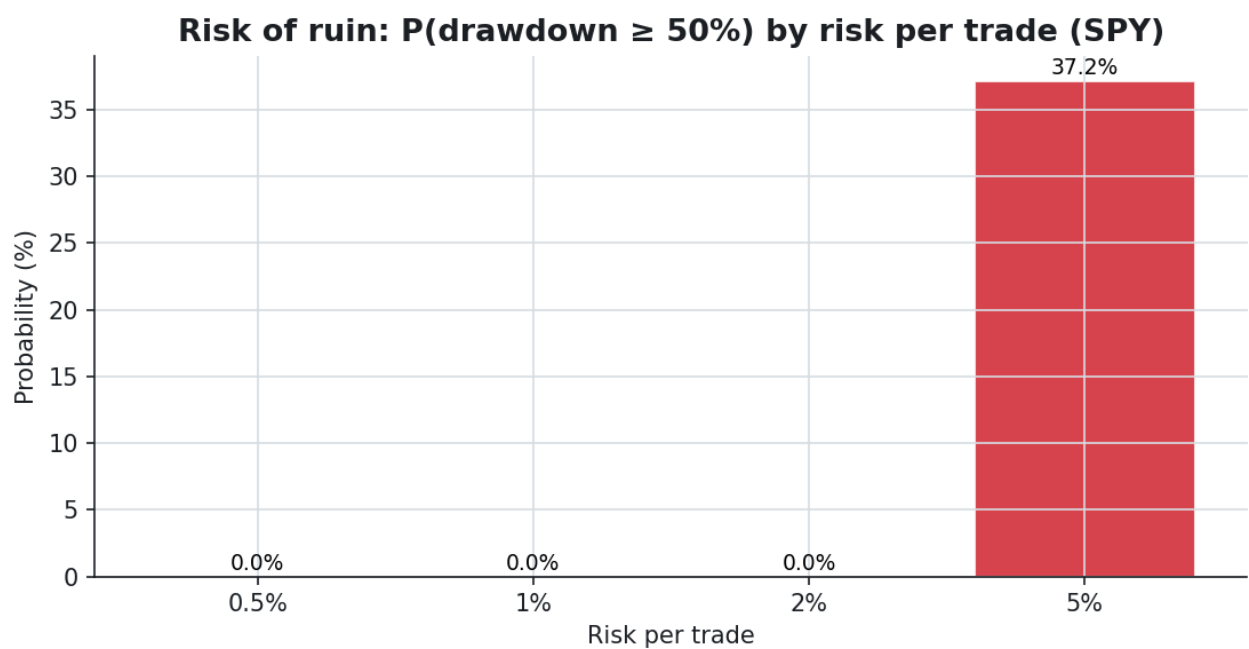


Monte-Carlo fan: median path and 5–95% band vs the single realized run.

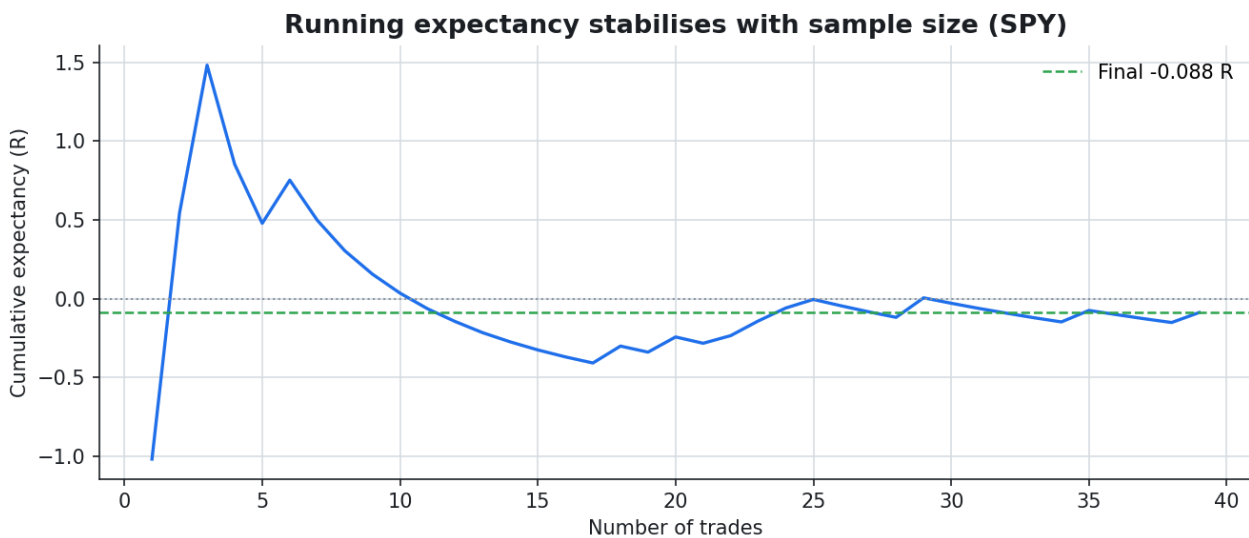


Distribution of final equity across the simulations.

3.2 Risk of ruin & sample size



Probability of a deep drawdown by risk per trade — it climbs non-linearly as the risk dial turns up.

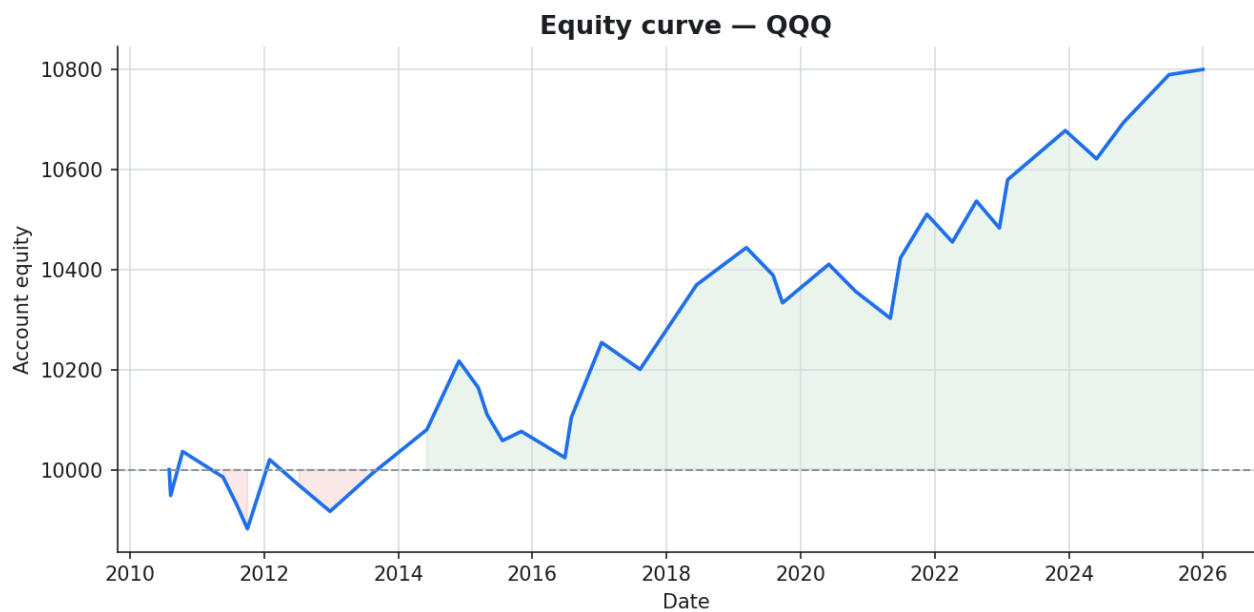


Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

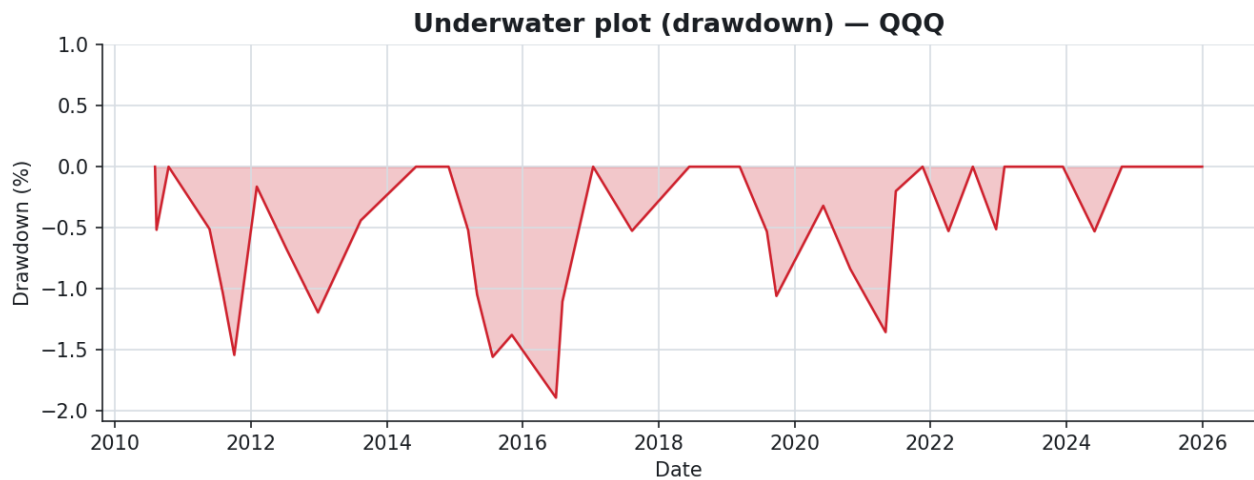
4. QQQ — the full fingerprint

Metric	Value	Metric	Value
Trades	37	Profit factor	1.83
Win rate	51.35%	Payoff (R)	1.74
Avg win	92.63 (+1.81R)	Avg loss	-53.31 (-1.04R)
Expectancy \$	21.63	Expectancy R	+0.422
Breakeven WR	36.54%	Actual WR	51.35%
Total return	+8.00%	CAGR	+0.50%
Max drawdown	1.89%	Sharpe / Sortino	0.42 / 20.79

■ Small sample (37 trades < 100): expectancy is still noise-dominated. Treat with skepticism.



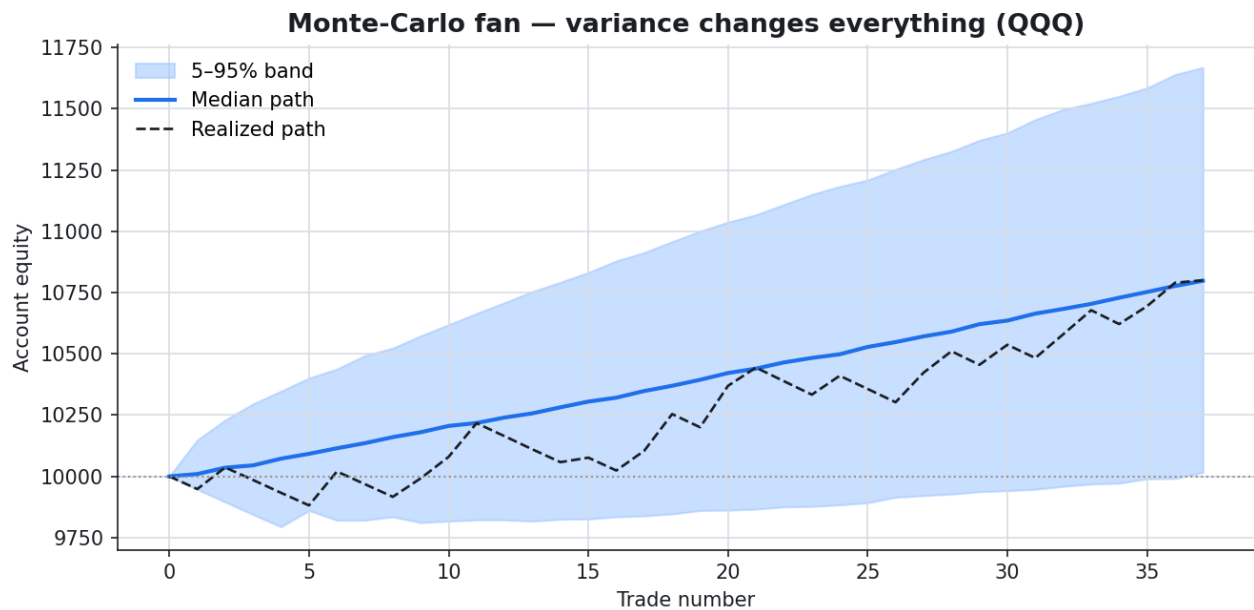
Realized equity curve, costs included.



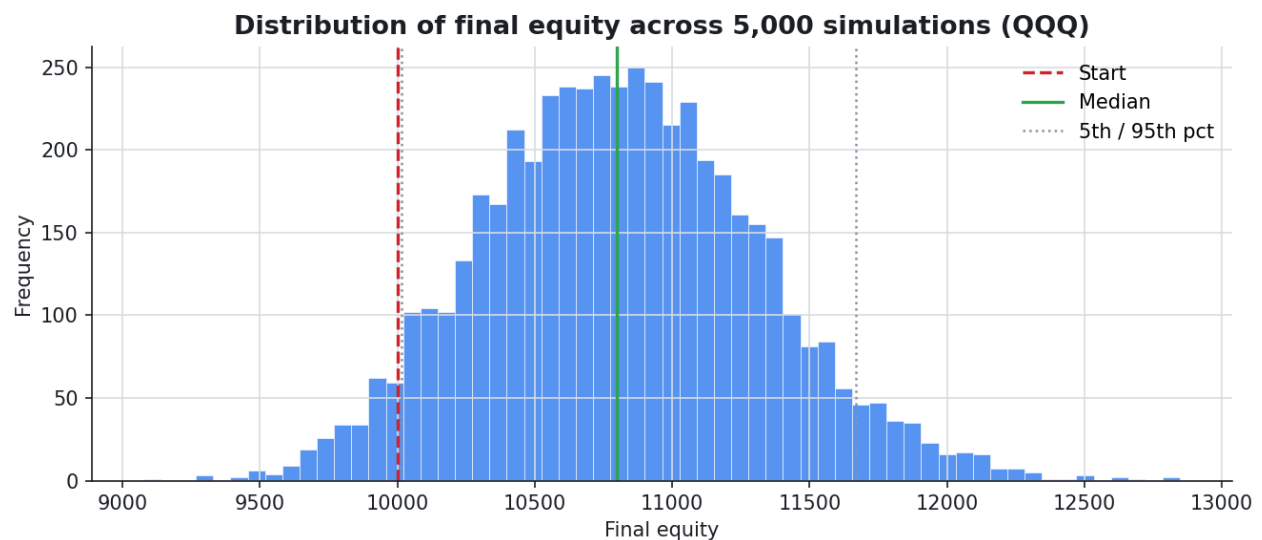
Underwater plot: time spent below the previous equity peak.

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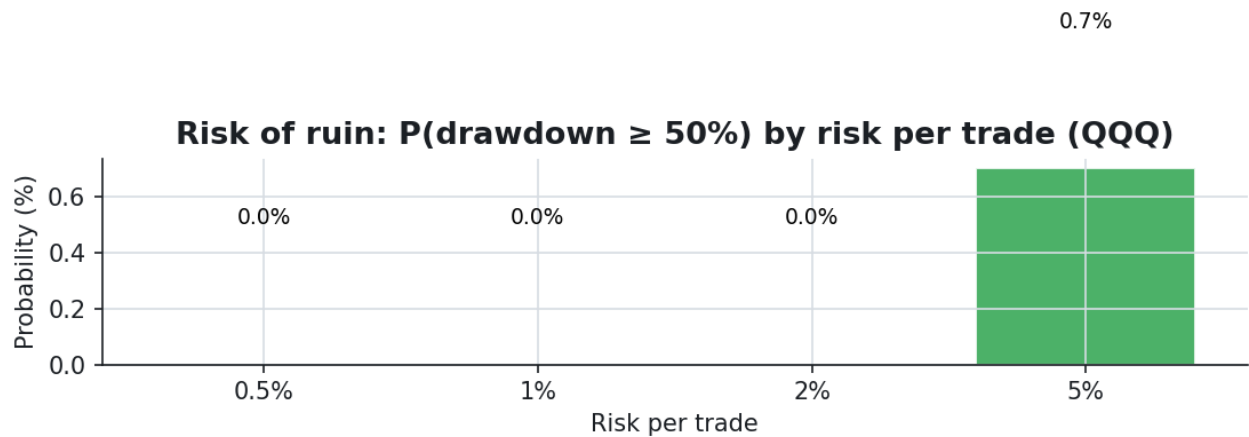


Monte-Carlo fan: median path and 5–95% band vs the single realized run.

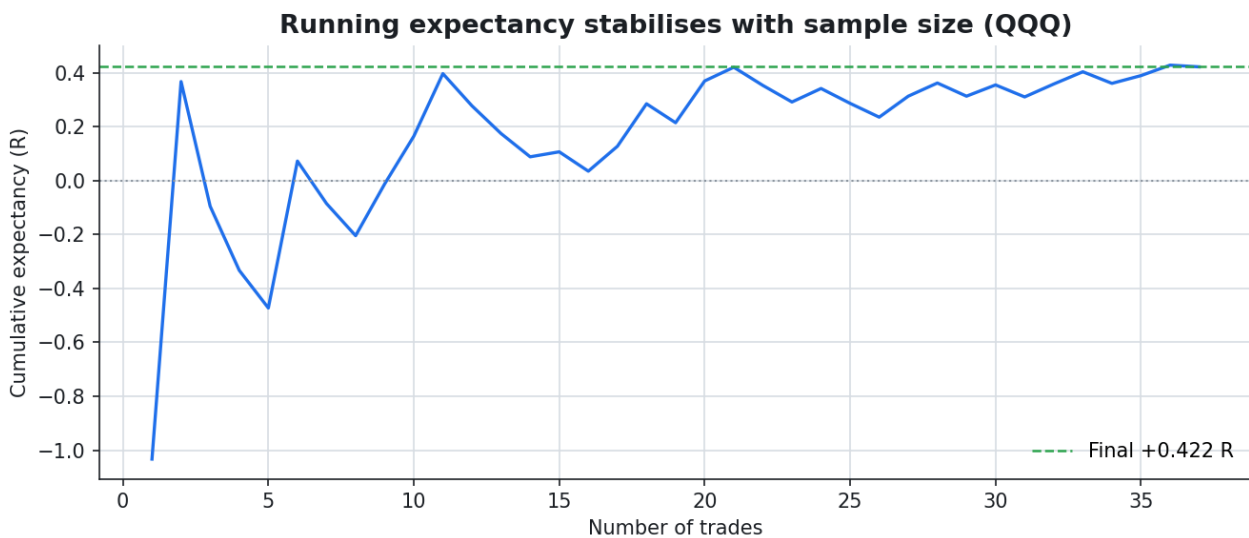


Distribution of final equity across the simulations.

4.2 Risk of ruin & sample size



Probability of a deep drawdown by risk per trade — it climbs non-linearly as the risk dial turns up.



Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

5. PETR4.SA — the full fingerprint

Metric	Value	Metric	Value
Trades	42	Profit factor	0.84
Win rate	35.71%	Payoff (R)	1.52
Avg win	80.71 (+1.63R)	Avg loss	-53.60 (-1.08R)
Expectancy \$	-5.63	Expectancy R	-0.109
Breakeven WR	39.75%	Actual WR	35.71%
Total return	-2.37%	CAGR	-0.16%
Max drawdown	3.68%	Sharpe / Sortino	-0.14 / -4.86

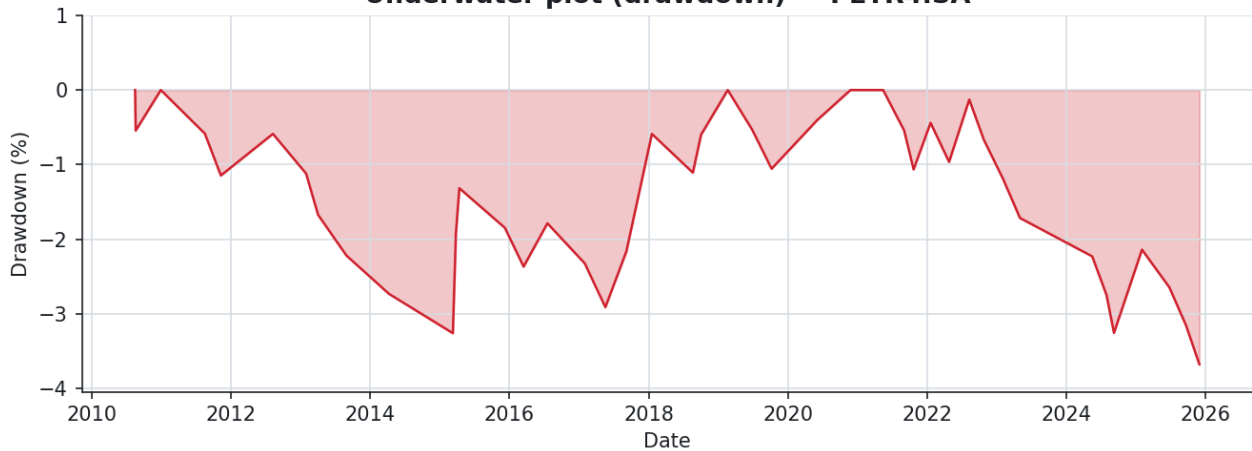
■ Small sample (42 trades < 100): expectancy is still noise-dominated. Treat with skepticism.

Equity curve — PETR4.SA



Realized equity curve, costs included.

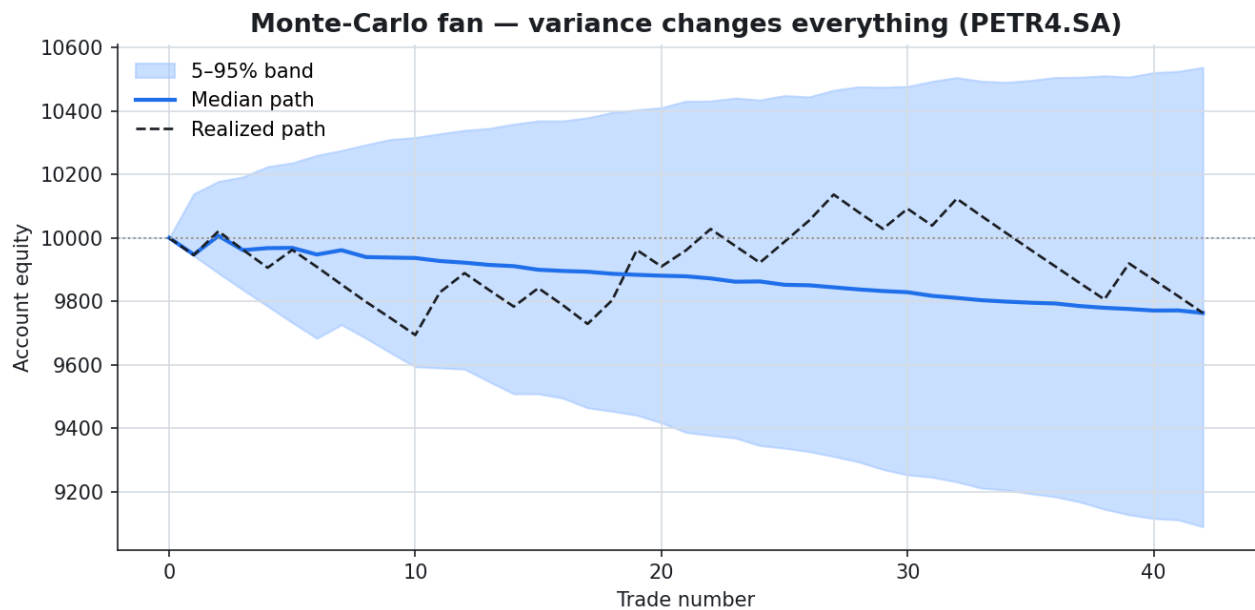
Underwater plot (drawdown) — PETR4.SA



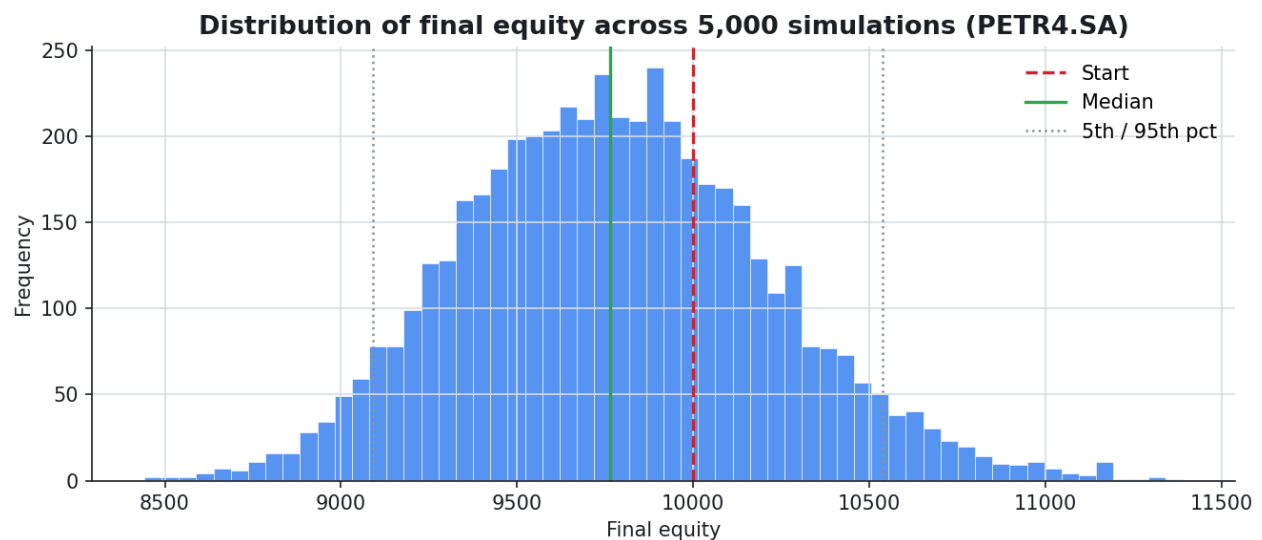
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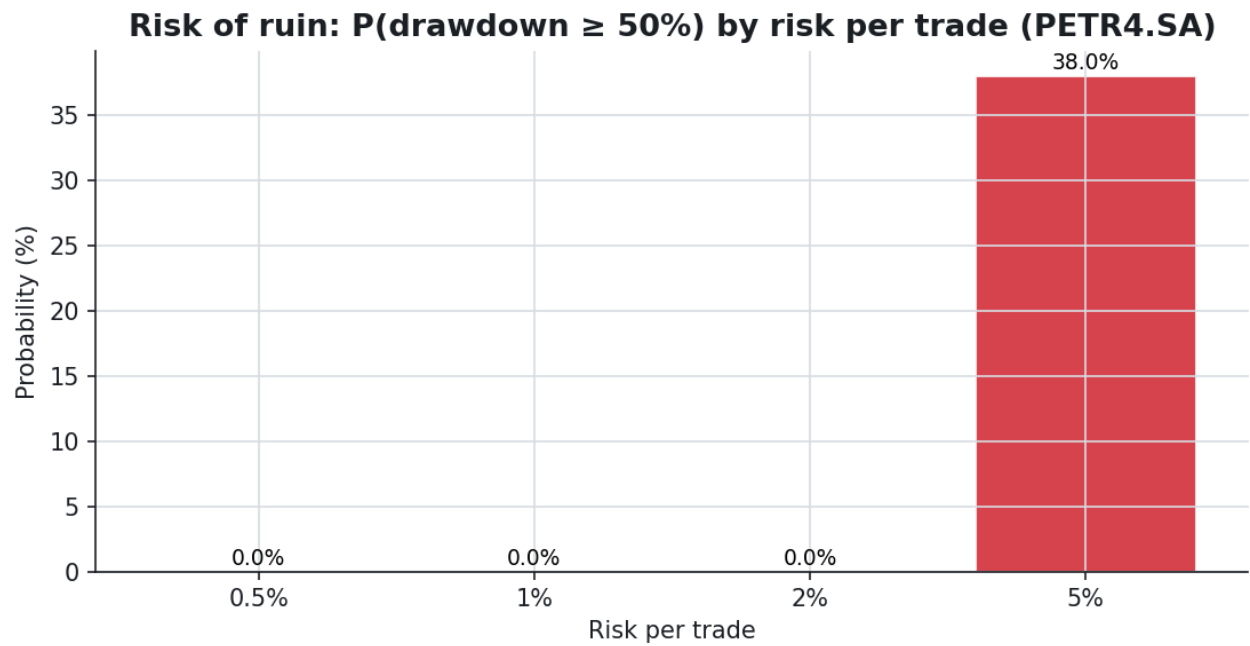


Monte-Carlo fan: median path and 5–95% band vs the single realized run.

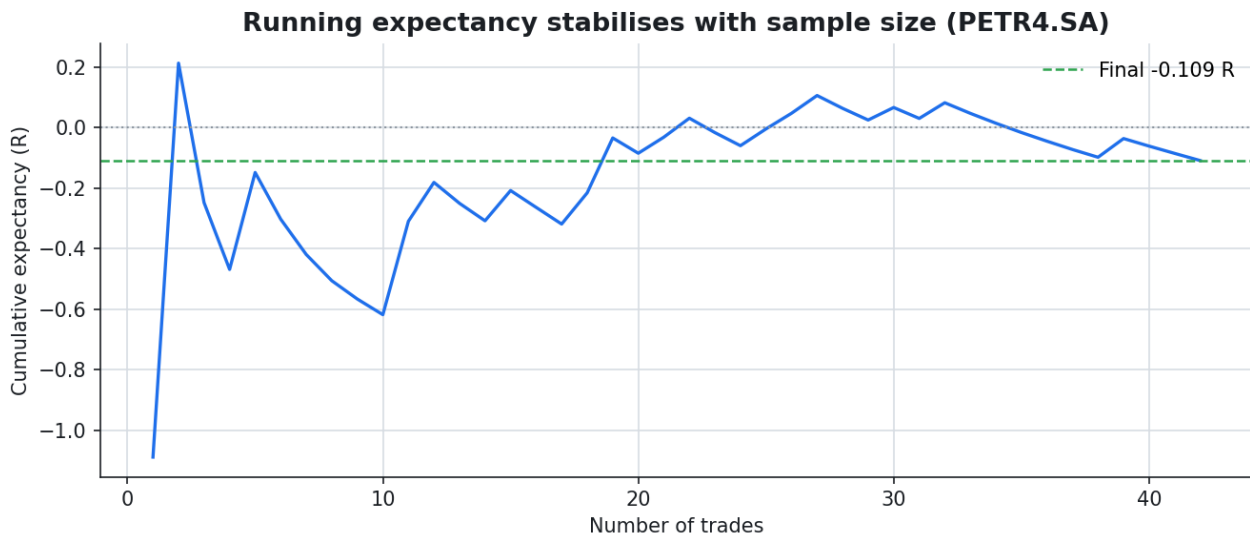


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Probability of a deep drawdown by risk per trade — it climbs non-linearly as the risk dial turns up.



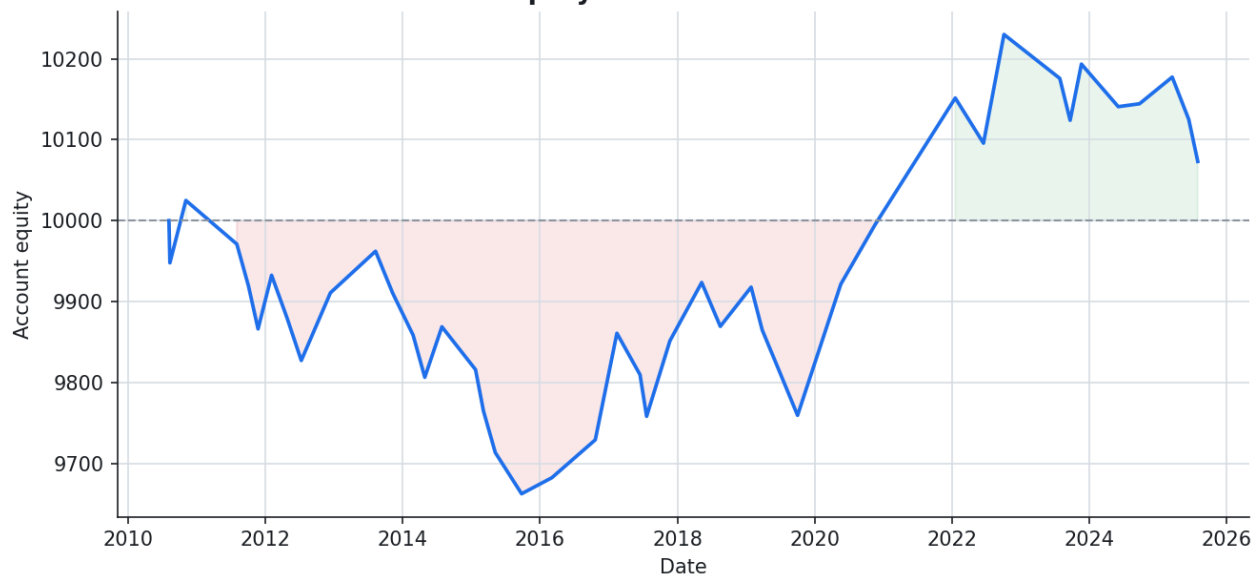
Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

6. VALE3.SA — the full fingerprint

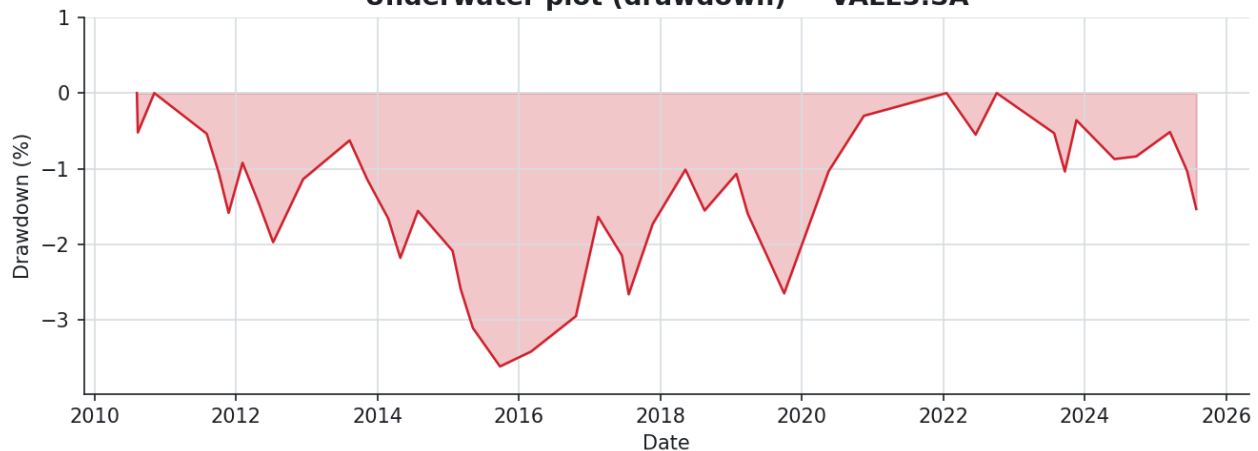
Metric	Value	Metric	Value
Trades	43	Profit factor	1.06
Win rate	41.86%	Payoff (R)	1.48
Avg win	77.00 (+1.56R)	Avg loss	-52.51 (-1.06R)
Expectancy \$	1.70	Expectancy R	+0.039
Breakeven WR	40.37%	Actual WR	41.86%
Total return	+0.73%	CAGR	+0.05%
Max drawdown	3.62%	Sharpe / Sortino	0.04 / 2.40

■ Small sample (43 trades < 100): expectancy is still noise-dominated. Treat with skepticism.

Equity curve — VALE3.SA

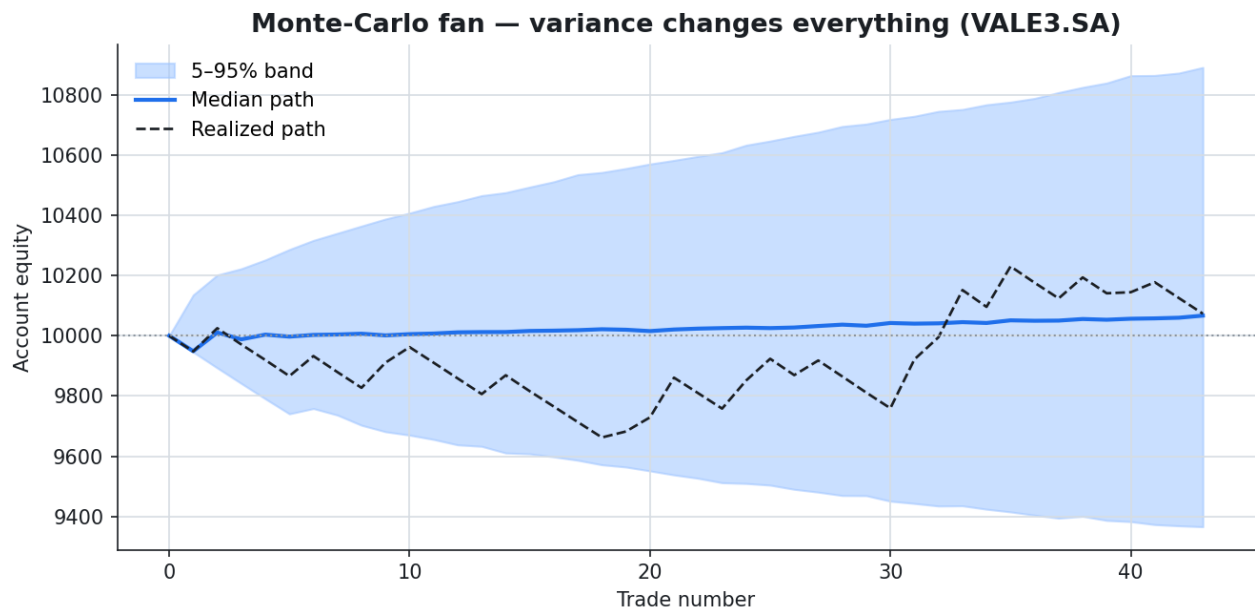


Underwater plot (drawdown) — VALE3.SA

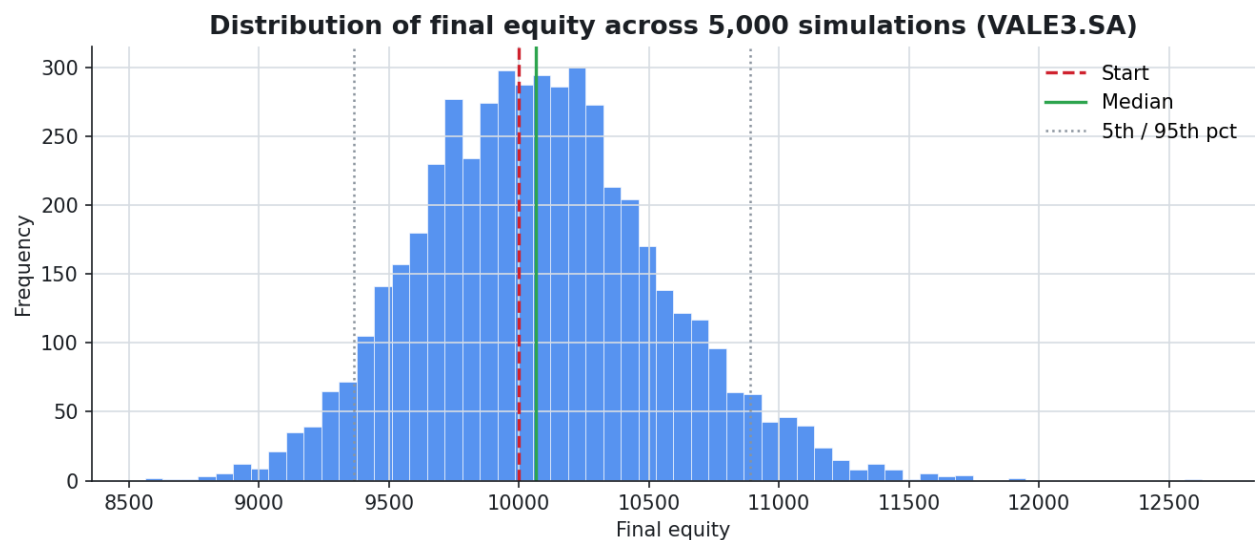


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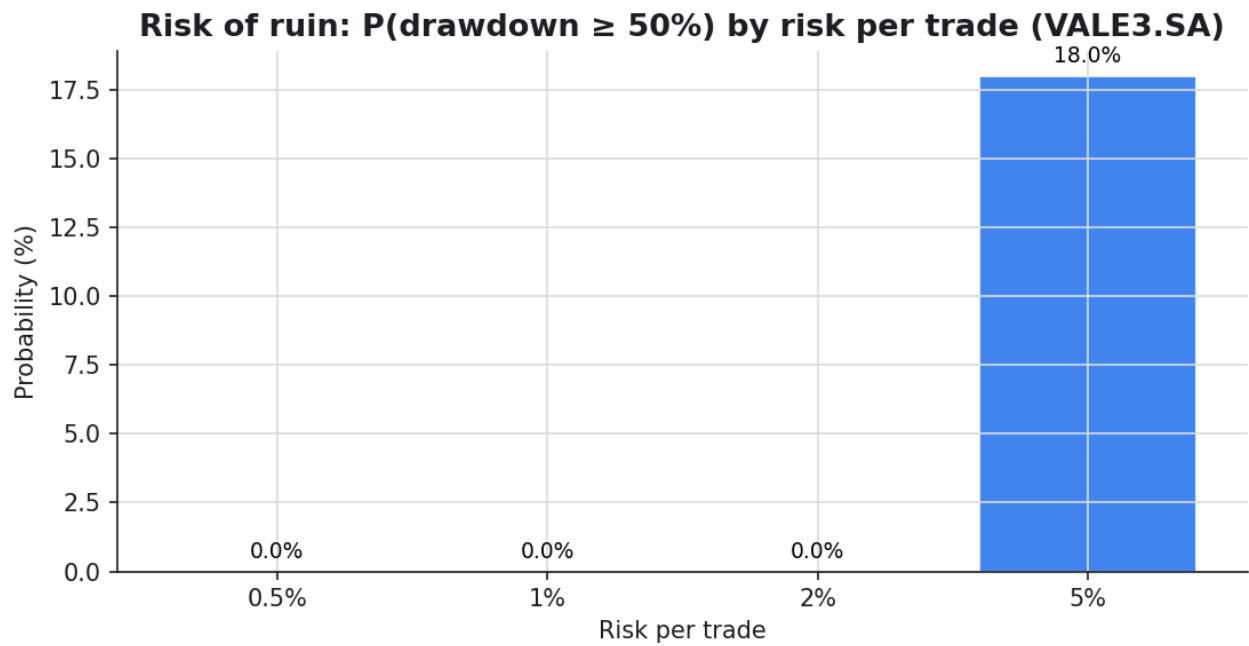


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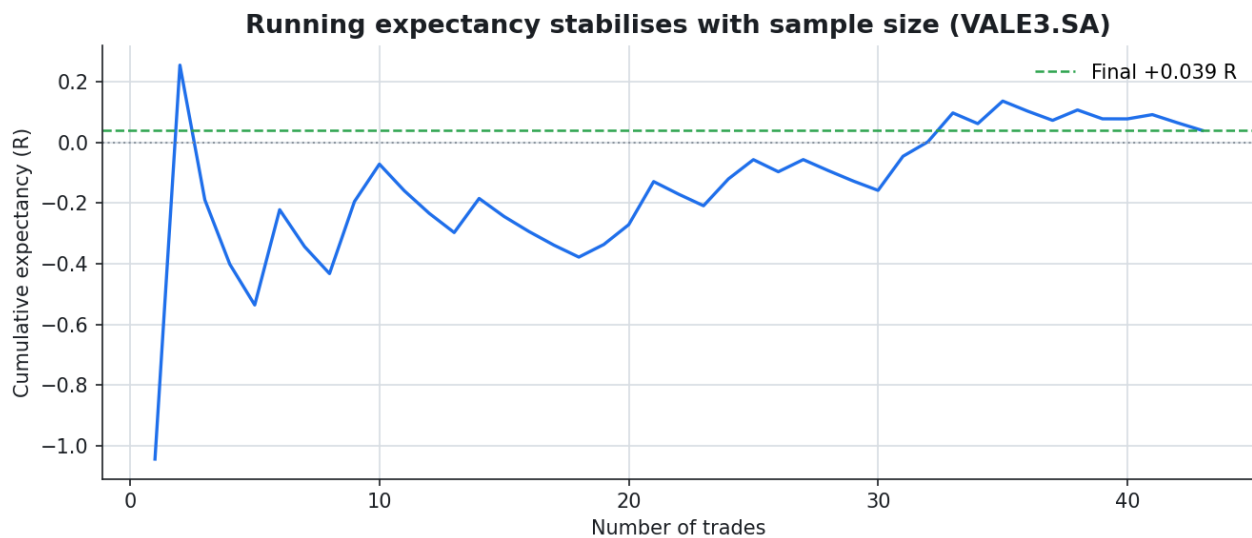


Distribution of final equity across the simulations.

6.2 Risk of ruin & sample size



Probability of a deep drawdown by risk per trade — it climbs non-linearly as the risk dial turns up.



Running expectancy vs trade count: one trade means nothing; the average only settles with sample size.

7. ITUB4.SA — the full fingerprint

Metric	Value	Metric	Value
Trades	40	Profit factor	1.40
Win rate	47.50%	Payoff (R)	1.54
Avg win	82.28 (+1.65R)	Avg loss	-53.33 (-1.07R)
Expectancy \$	11.09	Expectancy R	+0.222
Breakeven WR	39.32%	Actual WR	47.50%
Total return	+4.43%	CAGR	+0.28%
Max drawdown	4.70%	Sharpe / Sortino	0.25 / 10.54

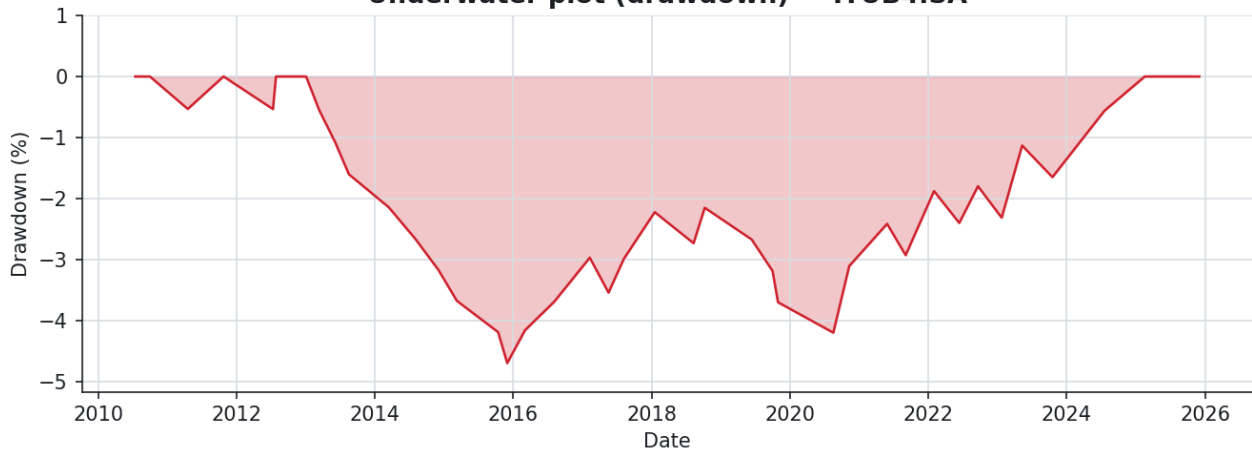
■ Small sample (40 trades < 100): expectancy is still noise-dominated. Treat with skepticism.

Equity curve — ITUB4.SA



Realized equity curve, costs included.

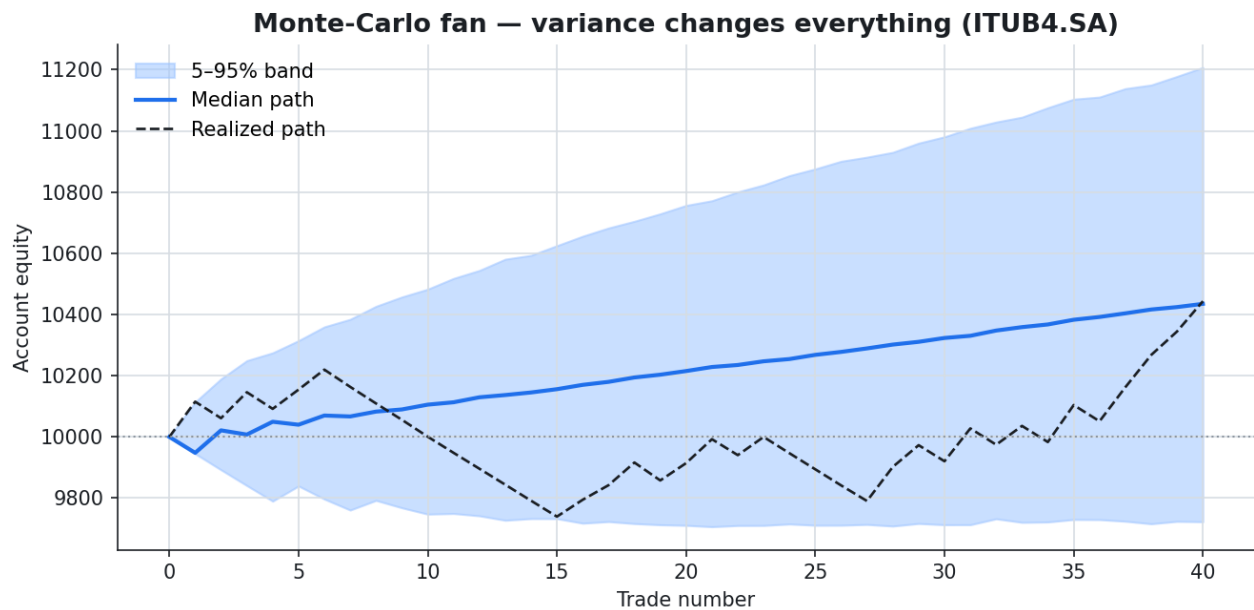
Underwater plot (drawdown) — ITUB4.SA



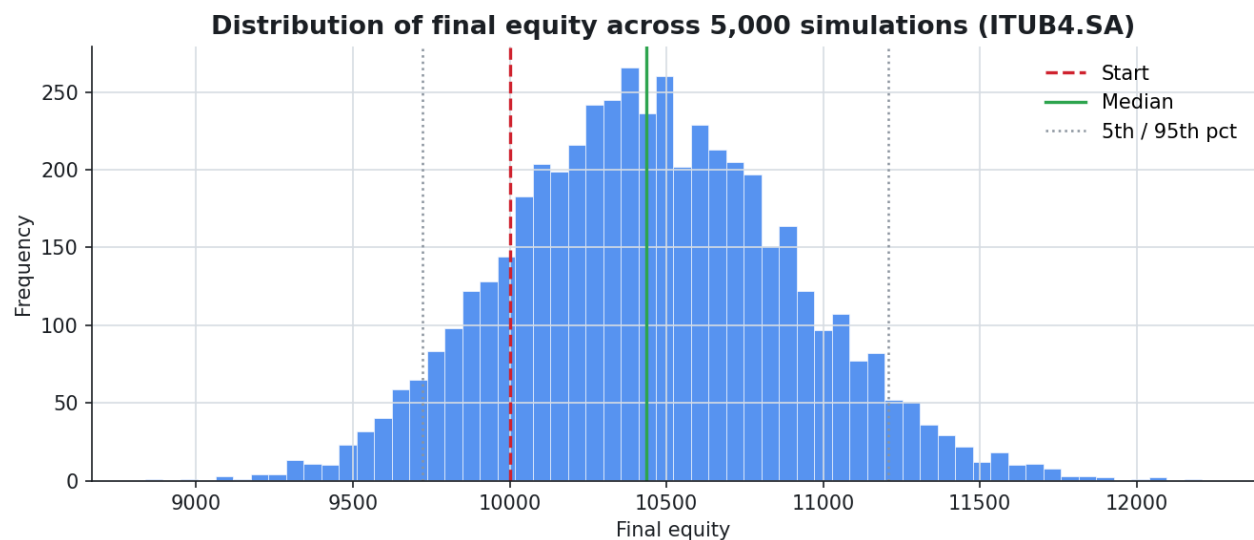
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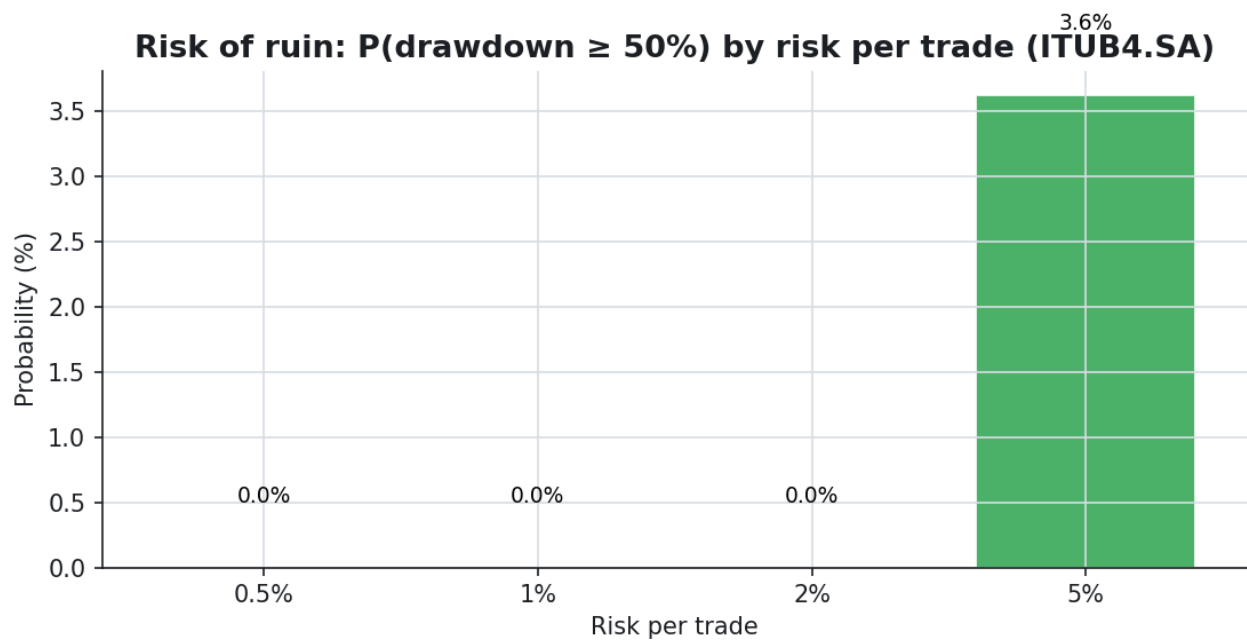


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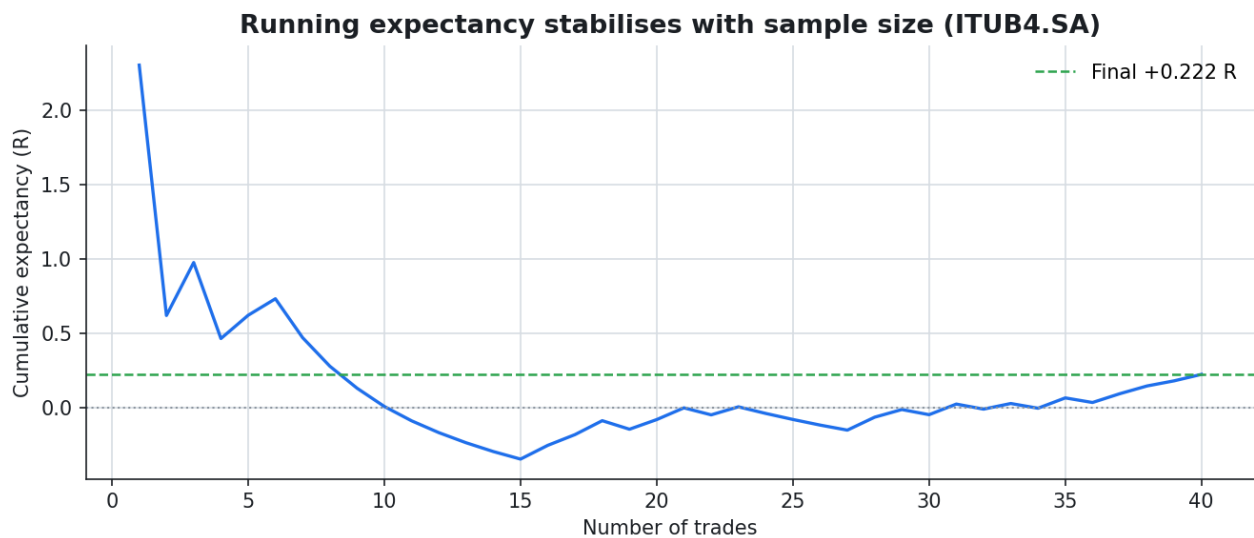


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8. The recovery math — why protecting capital wins

Losses and the gains needed to undo them are asymmetric. A 50% drawdown does not need 50% to recover — it needs 100%. This is why risk of ruin, not raw return, is the metric that keeps a trader in the game.

Drawdown suffered	Gain required to recover
10%	11%
20%	25%
30%	43%
40%	67%
50%	100%

Disclaimer

This is an educational and research project. Nothing here is investment advice. The example strategy is a didactic moving-average crossover, not a system with a guaranteed edge. The backtester *measures* an edge; it does not create one. Past performance does not guarantee future results.